

# **Process Transfer Plan**

**A-Mab Drug Substance — PTP-001**

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

# Table of contents

|                                                                |           |
|----------------------------------------------------------------|-----------|
| Approvals . . . . .                                            | 3         |
| Abbreviations . . . . .                                        | 3         |
| <b>1 Purpose and scope</b>                                     | <b>4</b>  |
| <b>2 Product and process description</b>                       | <b>6</b>  |
| <b>3 Sending and receiving sites</b>                           | <b>8</b>  |
| <b>4 Site equivalency analysis</b>                             | <b>9</b>  |
| <b>5 Scale-down model and comparability strategy</b>           | <b>11</b> |
| <b>6 Transfer of the process, methods and control strategy</b> | <b>13</b> |
| 6.1 Manufacturing process . . . . .                            | 13        |
| 6.2 Analytical methods . . . . .                               | 15        |
| 6.3 Control strategy . . . . .                                 | 16        |
| <b>7 Process performance qualification strategy</b>            | <b>18</b> |
| <b>8 Gap analysis</b>                                          | <b>19</b> |
| <b>9 Roles and responsibilities</b>                            | <b>22</b> |
| <b>10 Deliverables and schedule</b>                            | <b>23</b> |
| <b>11 Approvals</b>                                            | <b>24</b> |
| <b>References</b>                                              | <b>25</b> |

| Field          | Value                                        |
|----------------|----------------------------------------------|
| Document ID    | PTP-001                                      |
| Document class | Process Transfer Plan                        |
| Title          | Process Transfer Plan — A-Mab Drug Substance |
| Product        | A-Mab                                        |
| Version        | 1.0                                          |
| Effective date | 2026-07-24                                   |
| Status         | Approved (synthetic)                         |

| Field           | Value                                                                                                                      |
|-----------------|----------------------------------------------------------------------------------------------------------------------------|
| Document owner  | Manufacturing Science & Technology (MSAT)                                                                                  |
| Sponsor / sites | Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development)<br>→ Novacyte Biologics — Grafton, WI (Commercial DS) |

**SYNTHETIC DOCUMENT — NOT A REAL RECORD.** Illustrative content generated from a seeded model of the *A-Mab* case study (CMC Biotech Working Group, 2009). Novacyte Biologics and all sites, SOP/AMV numbers and signatories are fictional. For NLP corpus development only; not for any regulated use.

## Approvals

Table 1: Approval record (synthetic; signatures not applied).

| Role     | Function                   | Name (synthetic) | Signature | Date |
|----------|----------------------------|------------------|-----------|------|
| Author   | Process Development / MSAT | —                | —         | —    |
| Reviewer | Analytical Development     | —                | —         | —    |
| Reviewer | Manufacturing Science      | —                | —         | —    |
| Reviewer | Statistics                 | —                | —         | —    |
| Approver | Quality Assurance          | —                | —         | —    |

## Abbreviations

2-AB, 2-aminobenzamide; AMV, analytical method validation; CEX, cation exchange; CQA, critical quality attribute; DNA, deoxyribonucleic acid; ELISA, enzyme-linked immunosorbent assay; HCP, host cell protein; HILIC-UPLC, hydrophilic interaction ultra-performance liquid chromatography; HMW, high molecular weight; ICH, International Council for Harmonisation; icIEF, imaged capillary isoelectric focusing; IgG, immunoglobulin G; LOQ, limit of quantitation; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; NTU, nephelometric turbidity unit; PD, Process Development; qPCR, quantitative polymerase chain reaction; RSD, relative standard deviation; SEC-HPLC, size-exclusion high-performance liquid chromatography; SOP, standard operating procedure; TCID<sub>50</sub>, median tissue culture infectious dose; TFF, tangential flow filtration; UV, ultraviolet; XMuLV, xenotropic murine leukaemia virus.

# 1 Purpose and scope

Novacyte Biologics is transferring the A-Mab drug substance process from Novacyte Biologics — Cambridge, MA (Development) to Novacyte Biologics — Grafton, WI (Commercial DS), and this plan defines the scope of that transfer, the activities it comprises, and the deliverables that must be complete before process performance qualification can start at the receiving site. The process moves as a whole. No unit operation is redeveloped at the receiving site, and no set-point changes by the act of being transferred.

Technology transfer is an element of the pharmaceutical quality system, and it is expected to deliver the knowledge that allows the receiving site to run the process as it was intended to run (International Council for Harmonisation 2008). It is also the point at which the Stage 1 process design work is consolidated for a commercial process (U.S. Food and Drug Administration 2011). The receiving site has to operate inside ranges that data support, so the transfer and the process characterization campaign are planned as one programme. This plan is the parent of that campaign and states why each of its documents exists.

The package contains 20 controlled documents, and Table 1.1 lists all of them except this plan. A pre-characterization risk assessment (RA-001) fixes the scope of the studies, a master plan (PCMP-001) sets the approach that every study shares, a plan and a report cover each unit operation (PCP-003 through PCP-010, PCR-003 through PCR-010), and a master report (PCMR-001) consolidates the outcome for the process as a whole. Each of those documents is a deliverable of this transfer, and the sequence in which they are produced is given in §10.

Table 1.1: The A-Mab document package that this transfer plan parents.

| Document ID | Document class                               | Subject                            |
|-------------|----------------------------------------------|------------------------------------|
| RA-001      | Pre-Characterization Process Risk Assessment | A-Mab Drug Substance               |
| PCMP-001    | Process Characterization Master Plan         | A-Mab Drug Substance               |
| PCP-003     | Process Characterization Plan                | Production Bioreactor (Step 3)     |
| PCR-003     | Process Characterization Report              | Production Bioreactor (Step 3)     |
| PCP-004     | Process Characterization Plan                | Harvest and Clarification (Step 4) |
| PCR-004     | Process Characterization Report              | Harvest and Clarification (Step 4) |
| PCP-005     | Process Characterization Plan                | Protein A Chromatography (Step 5)  |
| PCR-005     | Process Characterization Report              | Protein A Chromatography (Step 5)  |

| Document ID | Document class                         | Subject                                   |
|-------------|----------------------------------------|-------------------------------------------|
| PCP-006     | Process Characterization Plan          | Low-pH Viral Inactivation (Step 6)        |
| PCR-006     | Process Characterization Report        | Low-pH Viral Inactivation (Step 6)        |
| PCP-007     | Process Characterization Plan          | Cation Exchange Chromatography (Step 7)   |
| PCR-007     | Process Characterization Report        | Cation Exchange Chromatography (Step 7)   |
| PCP-008     | Process Characterization Plan          | Anion Exchange Chromatography (Step 8)    |
| PCR-008     | Process Characterization Report        | Anion Exchange Chromatography (Step 8)    |
| PCP-009     | Process Characterization Plan          | Small-Virus Retentive Filtration (Step 9) |
| PCR-009     | Process Characterization Report        | Small-Virus Retentive Filtration (Step 9) |
| PCP-010     | Process Characterization Plan          | Ultrafiltration / Diafiltration (Step 10) |
| PCR-010     | Process Characterization Report        | Ultrafiltration / Diafiltration (Step 10) |
| PCMR-001    | Process Characterization Master Report | A-Mab Drug Substance                      |

The scope of this plan is the drug substance process, Steps 3 to 10, as it will be operated at the receiving site. Three things sit outside it. Drug product manufacture and fill-finish are not covered, and neither is cell bank manufacture and testing. The execution of process performance qualification is not covered either, although §7 states what that stage requires and when it may begin, because the readiness decision belongs to this plan even though the qualification itself does not.

## 2 Product and process description

A-Mab is a humanized IgG1 monoclonal antibody produced in mammalian cell culture, and its main mechanism of action is antibody-dependent cellular cytotoxicity, so the glycan attributes that modulate effector function carry an efficacy risk and are treated as quality attributes throughout the process (CMC Biotech Working Group 2009). The purification train is the platform train that has already supported three products of the same class (X-Mab, Y-Mab and Z-Mab). What moves to the receiving site is therefore a platform process, and the platform history is part of the knowledge that moves with it.

The drug substance process comprises 8 unit operations, numbered Steps 3 to 10, and it begins with a fed-batch culture at a working volume of 15,000 L. Purification follows the platform sequence of Protein A capture, low pH viral inactivation, cation exchange, anion exchange, small virus retentive filtration and ultrafiltration with diafiltration into the formulation buffer. The final step delivers drug substance at a target concentration of 75 g/L, and Table 2.1 lists every step with the role it holds in the control strategy.

Table 2.1: The A-Mab drug substance process train and the role of each step.

| Step | Unit operation                   | Principal role                                                               |
|------|----------------------------------|------------------------------------------------------------------------------|
| 3    | Production Bioreactor            | Forms the glycan, charge-variant and aggregate CQAs (design-space step)      |
| 4    | Harvest and Clarification        | Primary recovery and clarification; forms no product-quality CQA             |
| 5    | Protein A Chromatography         | Capture; sets leached Protein A; principal HCP and DNA clearance             |
| 6    | Low-pH Viral Inactivation        | Low-pH hold; sets the cumulative XMuLV (enveloped) clearance                 |
| 7    | Cation Exchange Chromatography   | Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance    |
| 8    | Anion Exchange Chromatography    | Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA |
| 9    | Small-Virus Retentive Filtration | Dedicated small-virus removal; principal MVM clearance                       |
| 10   | Ultrafiltration / Diafiltration  | Formulation / mass balance; delivers drug substance at target concentration  |

The bioreactor forms most of the attributes the process has to control, since 7 of the 10 attributes in the register are set there, and they include the glycans, the charge

variants, aggregate and the impurity burden that the purification train then clears. The downstream steps remove impurities and virus without changing the glycan distribution, and Protein A capture in particular sets one attribute of its own, leached Protein A, and is the principal clearance step for host cell protein (HCP) and DNA. Cation exchange is the principal aggregate polish. Anion exchange is a flow-through polish that clears XMuLV, host cell protein and DNA.

Viral safety is modular. Clearance comes from 3 steps whose mechanisms are independent of one another, namely inactivation at low pH (Step 6), adsorption of virus at anion exchange (Step 8), and size-based retention at the virus filter (Step 9). Each contribution is measured in its own study and the cumulative claim is consolidated in PCMR-001 (International Council for Harmonisation 2023a). No single step carries the whole claim.

Two steps neither form nor clear a quality attribute. Harvest and clarification (Step 4) recovers the culture, and the final ultrafiltration and diafiltration step (Step 10) concentrates the product and exchanges it into the formulation buffer. Both transfer and qualify like every other step, but their parameters are keyed to process performance and mass balance instead of to a quality attribute.

### 3 Sending and receiving sites

The sending site is Novacyte Biologics — Cambridge, MA (Development) and the receiving site is Novacyte Biologics — Grafton, WI (Commercial DS), and both operate under one pharmaceutical quality system, one change control system and one deviation management system (International Council for Harmonisation 2008). Table 3.1 gives the activities each site holds once the transfer is complete.

Table 3.1: Activities held by each site after transfer.

| Activity                                              | Held by                                                   |
|-------------------------------------------------------|-----------------------------------------------------------|
| Process development and optimization                  | Sending site                                              |
| Scale-down model operation and qualification          | Sending site                                              |
| Characterization studies and their reports            | Sending site                                              |
| Analytical method validation and reference materials  | Sending site                                              |
| Control strategy definition                           | Sending site, approved by Quality Assurance at both sites |
| Commercial manufacture of drug substance              | Receiving site                                            |
| Batch record execution and in-process control         | Receiving site                                            |
| Release and in-process testing, after method transfer | Receiving site quality control                            |
| Process performance qualification                     | Receiving site                                            |
| Change control after handover                         | Receiving site, with sending site review                  |

Development work does not stop at handover. The sending site keeps the scale-down models, the characterization data and the analytical reference materials, so that a question raised during qualification at the receiving site can be answered on the same model that generated the original data. The receiving site owns execution from the point at which the master batch record is approved, and it owns any subsequent change to the process, with the sending site as a named reviewer.



## 4 Site equivalency analysis

The receiving site is able to operate the process as it was developed, and that conclusion rests on a comparison across five dimensions, namely the equipment train, the scale, process control capability, utilities and materials, and analytical capability, with the quality system common to both sites. Table 4.1 gives the comparison, and two of the five dimensions differ in a way that affects the process, each of those differences carrying an action.

Table 4.1: Site equivalency comparison across the dimensions that affect the process.

| Dimension               | Sending site                                      | Receiving site                                        | Assessment                                                           |
|-------------------------|---------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------------|
| Equipment train         | Scale-down model of every unit operation          | Commercial train of the same class and sequence       | Equivalent in class and sequence                                     |
| Scale                   | Bench and pilot scale-down systems                | Commercial scale (\$2)                                | Differs by design; scale-dependent parameters matched on performance |
| Process control         | Development control systems and single-unit skids | Plant control system with recipe management           | Assessed per parameter against the range the process needs           |
| Utilities and materials | Compendial and vendor-qualified supply            | Same specifications, commercial supply chain          | Equivalent, subject to vendor qualification                          |
| Analytical capability   | Methods validated at the sending site (Table 6.1) | Quality control laboratory, methods to be transferred | Equivalent after method transfer (\$6.2)                             |
| Quality system          | One quality system and one change control system  | One quality system and one change control system      | Common to both sites                                                 |

Scale is the first difference and it is structural. The sending site holds scale-down models of every unit operation, and the receiving site holds the commercial train, whose production bioreactor works at 15,000 L. Parameters that are independent of scale carry across unchanged, because they are properties of the chemistry and not of the vessel. Culture pH, temperature, protein load, conductivity, elution pH and hold time mean the same thing in a bench model as they do in a commercial

vessel. Parameters that depend on the geometry of the vessel or the column do not carry across in the same way, so they are matched on the performance they produce instead of on the value set at the panel. Each scale-down model qualification records that matching (SOP-1001), and the per-step characterization plans (PCP-003 through PCP-010) describe it for the step they cover.

Control capability is the second difference, and it is the one that decides how tightly a parameter has to be held. A range that is wide relative to what the equipment can hold is a different proposition from a range the equipment can only just achieve, and the classification scheme used across the campaign takes that into account (SOP-4001). Each classification is therefore made against the control capability of the receiving site's equipment, and not against what a development rig can hold. Capability is measured, not assumed. The receiving site supplies control capability data for every parameter in Table 5.1, and those data are an input to the classification recorded in each characterization report.

Utilities and raw materials were compared on grade and on supply, not on supplier identity. The utilities the process depends on (water, process gases, clean steam) and the consumables it depends on (buffers, media, single-use components, chromatography resins) meet the same specifications at both sites. Materials are bought from the same vendors wherever the vendor is qualified for commercial supply. Where a vendor is not qualified, the substitution is handled as a raw material change under change control, together with an assessment of the attribute that the material touches.

Analytical capability is equivalent in scope and is not yet in place. The methods the process needs are validated at the sending site, and Table 6.1 is the register of those validations. Each of them has to be transferred to the receiving site's quality control laboratory and requalified there before it can support a release decision or an in-process decision. The way that is done is set out in §6.2, and the residual risk is carried in the gap register (Table 8.1).

The comparison shows that the receiving site holds equipment of the same class in the same sequence, under the same quality system, and it fixes the control capability data the receiving site has to supply for every parameter in Table 5.1. It does not show that the equipment holds every parameter to the range the process needs, which is established parameter by parameter when those data are assessed in the characterization reports. It does not show that the process performs identically at the two sites either, and no comparison made on paper can show that. Equivalence of performance comes from the comparability data described in §5 and from the qualification batches described in §7. The design space itself is defined on qualified scale-down models and is not confirmed at commercial scale at the edges of its ranges.

## 5 Scale-down model and comparability strategy

The characterization that justifies the receiving site's operating ranges is executed on qualified scale-down models and not at commercial scale. That is the usual approach for a process of this kind, and it carries one condition. A scale-down model may support a range claim only where it has been qualified against data at scale for the step it represents (SOP-1001).

Qualification follows the same pattern at every step, in that variables which are independent of scale are operated at the commercial target values, and variables which depend on scale are set so that the small system reproduces the performance of the large one. The model is then run at the centre point in replicate, and its input and output attributes (titre, impurity levels, step yield, the quality attributes the step governs) are compared with data from the development and clinical campaigns at scale. Those replicate centre points also give the pure error term that the lack-of-fit test in each design of experiments study needs, so model qualification and study design draw on the same runs.

Table 5.1 gives the characterization scope for each step and names the documents that carry it. Of the 37 parameters in the process, 22 are studied in multivariate designs and 15 are assessed one factor at a time. The split is set by the pre-characterization risk assessment (RA-001) from the potential impact of a parameter on a quality attribute and its potential to interact with other parameters, and it is not revisited in this plan (International Council for Harmonisation 2023b). Two steps carry no multivariate study at all, harvest and clarification (Step 4) and ultrafiltration and diafiltration (Step 10), because neither of them forms or clears a quality attribute.

Table 5.1: Characterization scope by step, with the plan and report that cover it.

| Step | Unit operation                 | Parameters | Multivariate | Univariate | Documents         |
|------|--------------------------------|------------|--------------|------------|-------------------|
| 3    | Production Bioreactor          | 9          | 5            | 4          | PCP-003 / PCR-003 |
| 4    | Harvest and Clarification      | 3          | 0            | 3          | PCP-004 / PCR-004 |
| 5    | Protein A Chromatography       | 6          | 4            | 2          | PCP-005 / PCR-005 |
| 6    | Low-pH Viral Inactivation      | 4          | 3            | 1          | PCP-006 / PCR-006 |
| 7    | Cation Exchange Chromatography | 5          | 4            | 1          | PCP-007 / PCR-007 |

| Step | Unit operation                   | Parameters | Multivariate | Univariate | Documents         |
|------|----------------------------------|------------|--------------|------------|-------------------|
| 8    | Anion Exchange Chromatography    | 5          | 4            | 1          | PCP-008 / PCR-008 |
| 9    | Small-Virus Retentive Filtration | 2          | 2            | 0          | PCP-009 / PCR-009 |
| 10   | Ultrafiltration / Diafiltration  | 3          | 0            | 3          | PCP-010 / PCR-010 |

The plans that execute this scope on the scale-down models are PCP-003 through PCP-010, one for each step, and each names the model it uses and the qualification that supports it. Table 5.2 lists the instrumented systems the studies run on, and those systems are held under calibration and change control at the sending site, with their calibration and maintenance records forming part of the data package the receiving site receives.

Table 5.2: Instrumented scale-down systems used in the characterization studies.

| Equipment  | Description                                             |
|------------|---------------------------------------------------------|
| EQ-CHR-118 | temperature-controlled chromatography chamber           |
| EQ-BRX-205 | production scale-down bioreactor pCO <sub>2</sub> probe |
| EQ-TFF-142 | ultrafiltration / diafiltration TFF skid                |

Comparability between the sites is demonstrated on material and not on documents. Engineering runs at the receiving site are executed at the set-points, with every parameter inside its normal operating range, and the resulting in-process attributes, step yields and drug substance quality attributes are compared with the sending site's data and with the acceptance criteria in Table 6.3. The comparison is made step by step and then across the train, because an impurity that three steps clear in series cannot be assessed at any one of them alone. Where an attribute falls outside the expected distribution, the difference is investigated and closed in the receiving site's quality system before the qualification batches start.

## 6 Transfer of the process, methods and control strategy

Transfer runs in three streams, which are the manufacturing process, the analytical methods and the control strategy. Each stream has its own deliverables and its own approval, and none of them is complete until Quality Assurance at both sites has signed it. The streams are not independent and their sequence matters, because a method has to be in place at the receiving site before the engineering runs can be assessed, and the control strategy cannot be finalized until the characterization reports it draws on are approved.

### 6.1 Manufacturing process

The manufacturing process transfers as a document set and as a training obligation. The sending site issues the process description, the parameter set-points with their normal operating ranges, the in-process control points with their action limits, and the sampling plan for each of the 8 unit operations. The receiving site converts these into its own master batch records and into the recipe configuration held in the plant control system. Both sites then reconcile the batch record against the process description line by line, and Manufacturing Science at each site signs the reconciliation before any material is charged. Operator training is documented before execution.

The parameter set is what that reconciliation is checked against, and all 37 parameters appear in the batch record with their set-point and their normal operating range, whatever their classification, so that an excursion is recognized when it happens and not at batch review. Classification decides what an excursion means and what has to be done about it, and not whether the value is recorded at all.

Table 6.1 lists the procedures and method validation records cited across the package, all of which are sending site documents. Each of them needs an equivalent in the receiving site's system, and the mapping between the two sets is a deliverable of the transfer (§10).

Table 6.1: Controlled documents cited across the A-Mab characterization package.

| Refer-<br>ence | Title                              | Type |
|----------------|------------------------------------|------|
| SOP-<br>1001   | Qualification of Scale-Down Models | SOP  |

| Reference | Title                                                                                 | Type              |
|-----------|---------------------------------------------------------------------------------------|-------------------|
| SOP-1002  | Design, Execution and Statistical Analysis of DoE Studies                             | SOP               |
| SOP-2005  | Harvest by Continuous Disk-Stack Centrifugation                                       | SOP               |
| SOP-2006  | Clarification by Depth and Sterile Filtration                                         | SOP               |
| SOP-2007  | Operation of the Protein A Capture Chromatography Step                                | SOP               |
| SOP-2008  | Chromatography Resin Life-Cycle, Packing and Sanitization                             | SOP               |
| SOP-2009  | Operation of the Low-pH Viral Inactivation Step                                       | SOP               |
| SOP-2010  | Operation of the Cation-Exchange Polishing Chromatography Step                        | SOP               |
| SOP-2011  | Operation of the Anion-Exchange Polishing Chromatography Step                         | SOP               |
| SOP-2012  | Operation and Post-Use Integrity Testing of the Small-Virus Retentive Filtration Step | SOP               |
| SOP-2013  | Operation of the Ultrafiltration / Diafiltration (Formulation) Step                   | SOP               |
| SOP-4001  | Process-Parameter Classification and Control-Strategy Definition                      | SOP               |
| AMV-3011  | Size-Variants (SEC-HPLC)                                                              | Method validation |
| AMV-3012  | Host-Cell Protein ELISA                                                               | Method validation |
| AMV-3013  | Charge Variants (icIEF)                                                               | Method validation |
| AMV-3014  | Residual DNA (qPCR)                                                                   | Method validation |
| AMV-3015  | Turbidity by Nephelometry (NTU)                                                       | Method validation |
| AMV-3016  | Leached Protein A by ELISA (ppm)                                                      | Method validation |
| AMV-3017  | XMuvLV Infectivity Titre (TCID50)                                                     | Method validation |
| AMV-3018  | MVM Infectivity Titre (TCID50/qPCR)                                                   | Method validation |
| AMV-3019  | Protein Concentration by A280 (UV)                                                    | Method validation |

## 6.2 Analytical methods

Analytical method transfer sits on the critical path, because no batch is released and no comparability claim is made until the receiving site's laboratory can run the methods that measure the attributes in question. Table 6.2 gives the validated performance of the methods for which performance records are held, which is the intermediate precision, the limit of quantitation, the accuracy and the share of the observed variability that the method itself contributes. The register of the validations themselves is Table 6.1.

Table 6.2: Validated performance records held for the analytical methods.

| MethodTitle                                             | Intermediate precision | LOQ                 | Accuracy (%) | Variance share |
|---------------------------------------------------------|------------------------|---------------------|--------------|----------------|
| AMV-3221 Aggregate by SEC-HPLC (verification-qualified) | 3.8 %RSD               | 0.2 % HMW           | 99.1         | 0.34           |
| AMV-3016 Leached Protein A by ELISA                     | 6.5 %RSD               | 0.25 ppm            | 97.4         | 0.41           |
| AMV-3017 XMuLV Infectivity Titre (TCID50)               | 0.3 log10 SD           | 1.5 log10 TCID50/mL | 96           | 0.28           |
| AMV-3019 Protein Concentration by A280 (UV)             | 1.2 %RSD               | 0.5 g/L             | 99.6         | 0.22           |
| AMV-3010 Released N-Glycan Map (2-AB HILIC-UPLC)        | 2.5 %RSD               | 0.5 % glycan        | 99           | 0.3            |
| AMV-3015 Turbidity by Nephelometry (NTU)                | 4 %RSD                 | 0.5 NTU             | 98           | 0.38           |
| AMV-3018 MVM Infectivity Titre (TCID50/qPCR)            | 0.32 log10 SD          | 1.4 log10 TCID50/mL | 96           | 0.3            |
| AMV-3012 Host-Cell Protein by ELISA                     | 9.5 %RSD               | 1 ng/mg             | 95           | 0.4            |

Acceptance criteria for each method transfer are set from those records, and the receiving site runs the method on split samples alongside the sending site, with the criterion for agreement derived from the intermediate precision recorded in the method's validation report. Two methods need more attention than the rest. AMV-3016 and AMV-3012 contribute the largest shares of the observed variability, at 0.41 and 0.40, and AMV-3012 is also the least precise of the methods that transfer, at 9.5 %RSD. Both measure impurities that the purification train exists to clear, so a difference between the two laboratories in either method would present as a difference in process performance.

Critical reagents are the practical risk in transferring an immunoassay. The HCP assay and the leached Protein A assay both depend on reagent lots that are not interchangeable without a bridging study. Reserve lots of both are held at the sending site until the receiving site has bridged its own, and that bridging is part of the method transfer protocol (GAP-02 in §8).

Not every method transfers. The infectivity assays for the two model viruses (XMuLV and MVM) support the clearance studies, and viral clearance is not measured on commercial batches at all. Clearance is claimed from spiking studies executed on qualified small-scale models, and the commercial process is never spiked (International Council for Harmonisation 2023a). The receiving site therefore transfers the release and in-process methods, and takes the viral clearance claim from the characterization package.

## 6.3 Control strategy

The control strategy is what the other two streams exist to deliver, and it has four elements, which are the parameter set-points and their ranges, the in-process controls with their action limits, release testing against the drug substance specification, and life-cycle controls such as resin and membrane reuse limits (International Council for Harmonisation 2012). The receiving site implements all four, and Quality Assurance at both sites approves the whole before the qualification batches begin.

Table 6.3 lists the attributes the control strategy exists to control, with the acceptance criterion and the criticality of each. The register holds 10 attributes, of which 6 are of high or very high criticality, and the 3 of very high criticality are high mannose and the two viral clearance claims. Criticality is treated as a continuum and not as a binary label (International Council for Harmonisation 2023b), so the attributes at the lower end of the register stay in the control strategy and are reported, with the depth of control proportionate to criticality.

Table 6.3: Drug substance quality attributes, acceptance criteria and criticality.

| CQA                                  | Category         | Acceptance                 | Criticality | Tool #1 |
|--------------------------------------|------------------|----------------------------|-------------|---------|
| Afucosylation                        | glycosylation    | 2-13 % of glycans          | H           | 60      |
| Galactosylation (total %Gal)         | glycosylation    | 10-40 % (G1+G2 equiv.)     | H           | 48      |
| High Mannose                         | glycosylation    | 3-10 % of glycans          | VH          | 80      |
| Aggregates (HMW)                     | size variant     | 0-5 % HMW (SEC)            | H           | 60      |
| Acidic charge variants (deamidation) | charge variant   | 18-40 % (CEX/icIEF)        | VL          | 4       |
| Host Cell Protein (HCP)              | process impurity | 0-100 ng/mg                | M-H         | 36      |
| Residual DNA                         | process impurity | 0-0.001 ng/dose            | VL          | 6       |
| Leached Protein A                    | process impurity | 0-5 ppm                    | L-M         | 16      |
| Viral clearance — XMuLV (enveloped)  | viral safety     | 16.7-30 log10 (cumulative) | VH          | 20      |



| CQA                                   | Category     | Acceptance                   | Criticality | Tool #1 |
|---------------------------------------|--------------|------------------------------|-------------|---------|
| Viral clearance — MVM<br>(parvovirus) | viral safety | 8.6-20 log10<br>(cumulative) | VH          | 20      |

Control is distributed across the train. The glycans are formed in the bioreactor and are not modified downstream, so their control is a culture control and is reported in PCR-003. Aggregate is formed upstream and polished at cation exchange (PCR-007), and host cell protein is cleared by three chromatography steps in series, so the acceptable output of any one of them depends on what the others achieve (PCR-005, PCR-007 and PCR-008). Viral clearance is credited from the three steps named in §2 and is consolidated in PCMR-001. No step is credited twice.

Until PCMR-001 is approved, the classifications carried in the batch record are provisional. The receiving site operates to the platform set-points and normal operating ranges in the interim, which is the conservative position, and revises the batch record when the campaign reports. This plan defines no design space and classifies no parameter, since both are outputs of the characterization package and neither is created by the act of transfer.

## 7 Process performance qualification strategy

Process performance qualification is Stage 2 of the lifecycle and it follows this transfer (U.S. Food and Drug Administration 2011). The receiving site manufactures 5 consecutive drug substance batches at commercial scale, at the set-points, with every parameter held inside its normal operating range. These batches are not experiments and no parameter is varied deliberately in them, and their purpose is to confirm that the process, as transferred and as controlled, delivers drug substance of the required quality reproducibly in the facility that will make it.

Qualification may start when the following are in place.

- The characterization reports for every unit operation are approved against the approach set in PCMP-001, and PCMR-001 states that the process is ready for Stage 2.
- Every method needed for in-process control and for release has been transferred to the receiving site and requalified there.
- Equipment, utility and facility qualification are complete, and the approved recipe is loaded in the plant control system.
- The engineering runs have been executed and the comparability assessment has been reviewed, with any deviation closed.
- The control strategy is approved at both sites and is reflected in the master batch record.
- The gaps named in §8 as prerequisites for qualification are closed, and each of the remaining gaps has an agreed closure plan.

Acceptance criteria for qualification are not set here, but in the qualification protocol, where they are derived from the design space and the process capability reported in PCMR-001 and are approved before the first batch is charged. Sampling during qualification is more extensive than routine sampling, and the additional sampling points are specified in the same protocol.

Stage 3 continued process verification follows qualification and sits outside this plan. The monitoring plan that supports it uses the same parameter set and the same methods, so it is written once PCMR-001 and the qualification report are available. Confirmation of the design space at commercial scale accumulates over the operating life of the process, because the qualification batches are executed at the set-points and say nothing about the edges of the ranges.

## 8 Gap analysis

Six gaps were identified during transfer planning, and none of them is closed at the date of this plan. Table 8.1 is the register, and each entry states the gap, its impact on the process or on the data, the action that closes it, and its status. Each of the six is discussed below, and four of them carry a residual risk that has to be explained.

Table 8.1: Transfer gap register, with the impact and the closing action for each gap.

| Gap    | Area              | Description                                                                | Impact                                                         | Mitigation                                                    | Status      |
|--------|-------------------|----------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------|-------------|
| GAP-01 | Materials         | Alternate Protein A resin has no platform bridging data                    | Second resin cannot enter commercial use on platform data      | Independent characterization per RA-004 before use            | Open        |
| GAP-02 | Analytical method | Critical reagent lots for the HCP and leached Protein A assays not bridged | A reagent lot difference can present as an impurity difference | Reagent bridging study during method transfer                 | In progress |
| GAP-03 | Validation        | Design space not confirmed at commercial scale at the edges of its ranges  | Operation near a range edge rests on model prediction          | Qualification inside the NOR; edge moves under change control | Open        |
| GAP-04 | Equipment         | Scale-dependent parameters cannot be matched setting for setting           | Small-scale performance may not reproduce at scale             | Performance matching per SOP-1001, tested in engineering runs | In progress |

| Gap    | Area             | Description                                                             | Impact                                                          | Mitigation                                                 | Status |
|--------|------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------|--------|
| GAP-05 | Documentation    | Procedures and method validations cited are sending site documents      | A batch record could cite a document the receiving site lacks   | Document mapping table issued with the process description | Open   |
| GAP-06 | Control strategy | Parameter classification is an output of the campaign, not yet complete | The batch record cannot yet state which parameters are critical | Interim operation to platform set-points and NORs          | Open   |

GAP-02, GAP-04 and GAP-06 close before qualification and are prerequisites for it. GAP-01 and GAP-05 constrain supply and documentation and do not block qualification, and GAP-03 is carried into Stage 3 and is not closed by qualification at all. Two of the six are recorded as in progress in Table 8.1 because their mitigations have already started, with reserve reagent lots held at the sending site (GAP-02) and the performance matching recorded as each scale-down model is qualified (GAP-04). Closure of GAP-01 and GAP-03 is owned by Process Development at the sending site, and closure of GAP-02 by Analytical Development at both sites. GAP-04 is owned by Manufacturing Science at both sites, and GAP-05 and GAP-06 by MSAT at the sending site. Quality Assurance decides in each case whether a gap is closed (§9).

The alternate Protein A resin cannot be qualified on platform data (GAP-01), because RA-004 assessed that resin and concluded that independent characterization is required, with no bridging from the platform data set. The receiving site therefore begins commercial manufacture with the primary resin alone, and the second resin enters the process only when its own characterization is complete and approved. What remains is a constraint on supply flexibility and not a risk to product quality, because the resin the characterization package covers is the resin the process will use.

The design space is defined on qualified scale-down models and is not confirmed at commercial scale at the edges of its ranges (GAP-03). Movement towards the edge of a range at the receiving site is supported by the fitted models and by the scale-down data, and not by experience at scale. Qualification does not close this gap, because the qualification batches are run at the set-points with every parameter inside its normal operating range. The gap closes as data at scale accumulate under continued process verification. A planned move towards the edge of a range before that point is assessed prospectively under change control, which is what movement inside a design space requires of the quality system in any case (International Council for Harmonisation 2009, 2008).

The two immunoassays are the transfer risk that surfaces last (GAP-02). HCP and leached Protein A are both measured by immunoassay, both depend on critical reagent lots, and both report on impurities the drug substance specification covers. AMV-3012 is the least precise of the methods that transfer, so a reagent difference between the two laboratories can look exactly like a process difference at the step that clears the impurity. The mitigation is a reagent bridging study during method transfer, with reserve lots held at the sending site until it is complete. Until that study is closed, an HCP result from the receiving site is compared with the sending site's data only alongside a split sample analysed at both.

Scale-dependent parameters are matched on the performance they produce and not setting for setting (GAP-04), which is a weaker warrant than a direct match. The geometry of a bench vessel and the geometry of a commercial vessel differ by construction, and a parameter that depends on that geometry cannot be held at the same value in both. The matching is recorded in each scale-down model qualification (SOP-1001) and is tested in the engineering runs, and that is why those runs are a prerequisite for qualification. Where an engineering run does not reproduce the scale-down performance for a step, the model for that step is requalified before its characterization data are used to support a range at commercial scale.

The last two gaps are documentary. The characterization package cites sending site procedures and method validation records while the receiving site holds its own set, so a mapping table is issued with the process description (GAP-05). Parameter classification is an output of the campaign, so the batch record cannot yet state which parameters are critical (GAP-06). Both gaps close on approval of the documents they depend on. Neither blocks the engineering runs.

## 9 Roles and responsibilities

Three functions own the transfer between them. Process Development and Manufacturing Science at the sending site own the process description, the characterization package and the control strategy. Manufacturing and Manufacturing Science at the receiving site own the batch records, the plant configuration, the engineering runs and execution. Quality Assurance at both sites owns approval, change control and the release decision, and it is the function that decides whether a gap is closed, with Table 9.1 giving the split by deliverable.

Table 9.1: Responsibility for each transfer deliverable.

| Deliverable                                      | Sending site | Receiving site | Quality Assurance |
|--------------------------------------------------|--------------|----------------|-------------------|
| Process description and parameter ranges         | Author       | Review         | Approve           |
| Characterization plans and reports               | Author       | Review         | Approve           |
| Scale-down model qualification records           | Author       | Review         | Approve           |
| Master batch records and recipe configuration    | Review       | Author         | Approve           |
| Analytical method transfer protocols and reports | Author       | Co-author      | Approve           |
| Engineering runs and comparability assessment    | Review       | Author         | Approve           |
| Control strategy                                 | Author       | Review         | Approve           |
| Qualification protocol and report                | Review       | Author         | Approve           |
| Gap register and gap closure                     | Co-owner     | Co-owner       | Approve           |

Where a deliverable has an author at one site and a reviewer at the other, both signatures are required. Neither site closes a transfer deliverable on its own.

## 10 Deliverables and schedule

The transfer is executed in five phases, each of which ends in a document that Quality Assurance approves, and Table 10.1 lists the deliverables in sequence. Calendar dates are held in the project schedule and are not reproduced here, because it is the sequence and not the date that governs. A phase is not complete until its deliverables are approved, and phases overlap where the dependency allows.

Table 10.1: Transfer deliverables and the phase that produces them.

| Phase              | Deliverables                                                                                                                                  | Owner                    |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Planning           | This transfer plan (PTP-001) and the pre-characterization risk assessment (RA-001)                                                            | Sending site MSAT        |
| Characterization   | Master plan (PCMP-001), the plan and report for each unit operation, master report (PCMR-001)                                                 | Sending site PD and MSAT |
| Transfer execution | Process description, master batch records, document mapping table, analytical method transfer reports, scale-down model qualification records | Both sites               |
| Readiness          | Engineering runs, comparability assessment, approved control strategy, gap closure record                                                     | Receiving site MSAT      |
| Qualification      | Qualification protocol, execution of the qualification batches, qualification report                                                          | Receiving site           |

The characterization phase is the long one, and it runs in parallel with the transfer execution phase for every step whose plan is already approved. The dependency that fixes the critical path is PCMR-001, because qualification cannot start until the master report is approved (§7).

# 11 Approvals

This plan takes effect when every signatory in Table 1 has approved it. Approval by Quality Assurance is required at both sites before any deliverable listed in Table 10.1 is executed. A change to this plan after approval is handled under change control, and a change that alters the scope stated in §1 requires re-approval by the same signatories.



## References

- CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.
- International Council for Harmonisation. 2008. *ICH Q10: Pharmaceutical Quality System*. ICH.
- International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.
- International Council for Harmonisation. 2012. *ICH Q11: Development and Manufacture of Drug Substances*. ICH.
- International Council for Harmonisation. 2023a. *ICH Q5A(R2): Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin*. ICH.
- International Council for Harmonisation. 2023b. *ICH Q9(R1): Quality Risk Management*. ICH.
- U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.